

AI: History, Agents and Search

CS21206: Foundations of AI and ML

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Agenda

- § Walking down the lanes of AI history
- § Intelligent agents and environments
- § Problem solving by search

Resources

- § Artificial Intelligence: A Modern Approach by Stuart Russell and Peter Norvig.
- § NPTEL Lectures on [An Introduction to Artificial Intelligence](#) by Prof. Mausam, IIT Delhi.
- § Stanford course on [Artificial Intelligence: Principles and Techniques \(CS221\)](#), Autumn 2025 by Prof. Percy Liang.
- § The slides are primarily inspired by these resources if not shamelessly copied.

The Dawn of Computing

§ 1945: The bomb and ENIAC



Image taken from: www.livescience.com

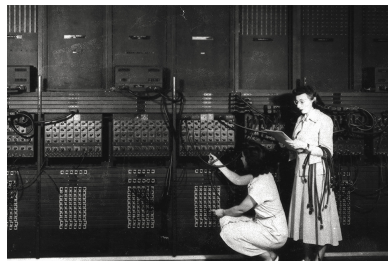


Image taken from: www.digitaltrends.com

Turing Asks the Question

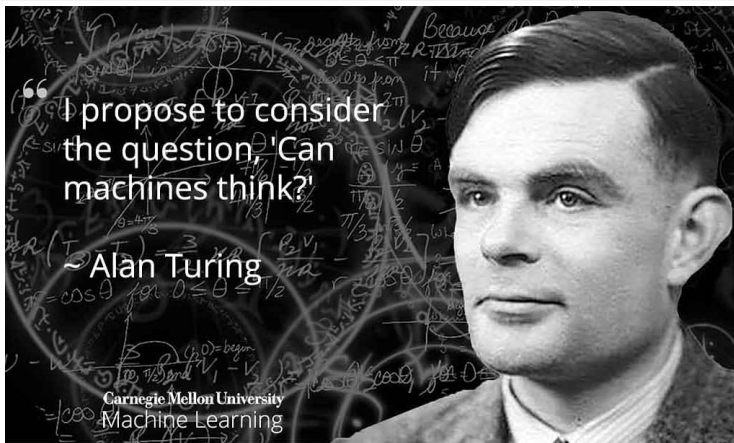


Image taken from: CMU Machine Learning Department, FB Page

- § 1950: Alan Turing's landmark paper, Computing Machinery and Intelligence.
- § This paper is remarkable not because it proposed any methods, but because it framed the philosophical discussions for decades to come.

The Turing Test

- § A machine is said to pass the *Turing test* if it can convince a human judge that it's actually a human through natural language dialogue.
- § Turing discusses two possible approaches.
 - ▶ The first is based on solving abstract problems like chess, which is the route taken by symbolic AI.
 - ▶ The second is where you build a machine and teach like a child, which is the route taken by neural and statistical AI.
- § History shows symbolic, neural and statistical routes dominated the AI landscape in different eras, sometimes coming back to life from oblivion.

The Birth of AI

- § 1956: John McCarthy organized workshop at Dartmouth College
- § McCarthy convinced Minsky, Claude Shannon, and Nathaniel Rochester to help him bring together U.S. researchers interested in the study of intelligence.
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- § Here was the first official usage of McCarthy's term artificial intelligence.
- § The proposal mentioned: *An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.*
- § The Dartmouth workshop did not lead to any new breakthroughs, but it did introduce all the major figures to each other and promoted why it was necessary for AI to become a separate field.

Early Successes

- § Checkers (1952): Samuel's program played at strong amateur level.
- § Problem solving (1955): Newell & Simon's Logic Theorist: prove theorems in Principia Mathematica.
- § This was followed up with the General Problem Solver, or GPS.
Within the limited class of puzzles, the program considered subgoals and possible action similar to that in which humans approached the same problems.
- § Herbert Gelernter (1959) constructed the Geometry Theorem Prover.

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- § Herbert Gelernter (1959) constructed the Geometry Theorem Prover.
- § Minsky supervised a series of students.
 - ▶ James Slagle's SAINT program (1963) was able to solve closed-form calculus integration problems typical of first-year college courses.
 - ▶ Tom Evans's ANALOGY program (1968) solved geometric analogy problems that appear in IQ tests.
 - ▶ Daniel Bobrow's STUDENT program (1967) solved algebra story problems.

Overwhelming Optimism

- § Machines will be capable, within twenty years, of doing any work a man can do. —Herbert Simon
- § Within 10 years the problems of artificial intelligence will be substantially solved. —Marvin Minsky
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Underwhelming Results!

The famous retranslation of “the spirit is willing but the flesh is weak” as “the vodka is good but the meat is rotten” illustrates the difficulties encountered.

- § 1966: ALPAC report cut off government funding for Machine Translation, first AI winter.

Implications of early era

Problems:

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Useful contributions (John McCarthy):

- § Lisp
- § Garbage collection
- § Time-sharing

Knowledge-based systems (70-80s)

Expert systems: elicit specific domain knowledge from experts in form of rules:

Example Rule

IF: 1) The stain of the organism is gramneg, and
2) The morphology of the organism is rod, and
3) The aerobicity of the organism is aerobic

THEN: There is strongly suggestive evidence that the class of the organism is enterobacteriaceae



Knowledge-based systems

Wins:

- § Knowledge helped both the **information** and **computation** gap
- § First real application that impacted industry

Problems:

- § Deterministic rules couldn't handle the **uncertainty** of the real world
- § Rules quickly became too **complex** to create and maintain

"In order to grasp any part, it is necessary to understand how it fits with other parts, presents a dense mass, with no easy footholds. Even having written the program, I find it near the limit of what I can keep in mind at once."

— Terry Winograd

1987: Collapse of Lisp machines and second AI winter

Artificial neural networks

§ Early history:

- ▶ **1943:** artificial neural networks (McCulloch/Pitts)
- ▶ **1949:** “cells that fire together wire together” (Hebb)
- ▶ **1958:** Perceptron algorithm (Rosenblatt)
- ▶ **1959:** ADALINE device (Widrow/Hoff)
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§ Revival of connectionism:

- ▶ **1980:** Neocognitron/CNNs (Fukushima)
- ▶ **1986:** Backpropagation (Rumelhardt, Hinton, Williams)
- ▶ **1989:** Handwritten digits for USPS (LeCun)

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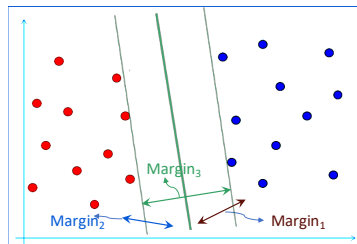
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- § (2016): AlphaGo uses deep reinforcement learning, defeats world champion Lee Sedol in Go

Statistical machine learning

- § In the middle, the field shifted focus toward mathematical rigor, temporarily moving away from “thinking machines” to “learning from data.”
- § **1985:** Bayesian networks (Pearl)
- § **1995:** Support vector machines (Cortes/Vapnik)

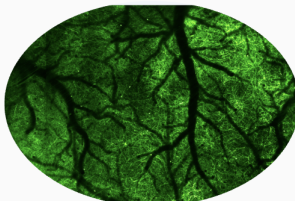


Three Intellectual Traditions



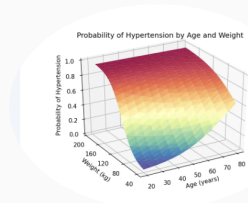
symbolic AI

Logic & Reasoning



neural AI

Neuroscience & Learning



statistical AI

Mathematics & Rigor

§ “AI War” between these traditions are over. Modern AI is a **synthesis**.

Images taken from: [wikipedia](https://en.wikipedia.org/wiki/File:Aristotle_Louvre_Museum.jpg), [news.berkeley.edu](https://www.berkeley.edu/news/news_article.php?id=123456), [blog.stata.com](https://www.stata.com/blog/2018/03/16/hypertension/)

Intelligent Agents

- § We begin by examining agents, environments and the coupling between them.
- § An **agent** is anything that can be viewed as perceiving its **environment** through **sensors** and acting upon that environment through **actuators**.

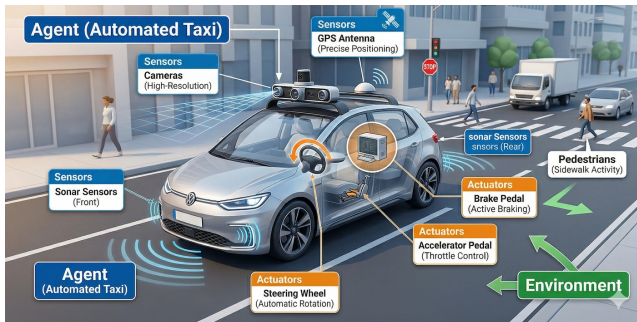


Image created by Gemini 3:

Agent-Environment Interaction

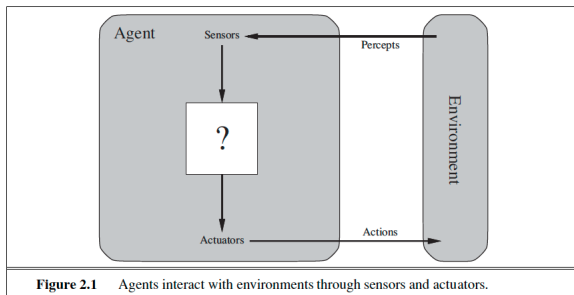


Image Credit: AIMA by Russell and Norvig

- § **Percept** refers to the agent's perceptual inputs at any given instant.
- § **Percept Sequence** is the complete history of everything the agent has ever perceived.
- § **Agent Function** maps any given percept sequence to an action.

The Vacuum-cleaner World

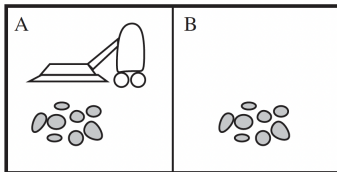


Image Credit: AIMA by Russell and Norvig

- § Just two locations: Squares A and B.
- § Vacuum agent **perceives** which square it is in and whether there is dirt in the square.
- § **Actions:** move left, move right, suck up the dirt, or do nothing.
- § **Agent function:** if the current square is dirty, then suck; otherwise, move to the other square.

A Rational Agent

- § A **rational agent** is one that does the right thing. But what does it mean to do the right thing?
- § Note that an agent's sequence of actions causes the environment to go through a sequence of states.
- § The notion of *right thing* is captured by **performance measure** that evaluates a given sequence of environment states.

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- § The notion of *right thing* is captured by **performance measure** that evaluates a given sequence of environment states.
- § Is the amount of dirt cleaned up in a single eight-hour shift a good performance measure?
- § An agent can maximize this performance measure by cleaning up the dirt, then dumping it all on the floor, then cleaning it up again and so on.
- § A more suitable performance measure would reward the agent for having a clean floor.
- § As a general rule, it is better to design performance measures according to what one actually wants in the environment.

Task Environments

§ Now, we must think about specifying **task environments**, which are essentially the “problems” to which rational agents are the “solutions”.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Image Credit: AIMA by Russell and Norvig

Properties of Task Environments

- § **Fully observable vs. partially observable:** If the sensors have access to the complete state of the environment (that are relevant to the choice of action) at each point in time, then it is fully observable.
- § An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data.
 - ▶ A vacuum agent with only a local dirt sensor cannot tell whether there is dirt in other squares.
 - ▶ An automated taxi cannot see what other drivers are thinking.

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 - ▶ An automated taxi cannot see what other drivers are thinking.
- § **Single agent vs. multiagent:** Even though this distinction is simple enough, there may be some subtle issues.
- § Does a taxi driver agent treat another vehicle as an agent, or can it be treated merely as part of the environment.
- § Such considerations give rise to **competitive** or **cooperative** multiagent task environments.

Properties of Task Environments

- § **Deterministic vs. stochastic:** If the next state of the environment is completely determined by the current state and the action executed by the agent, then we say the environment is deterministic; otherwise, it is stochastic.
- § The vacuum world is deterministic, but it can include stochastic elements such as randomly appearing dirt or unreliable suction *etc.*
- § “Stochastic” implies that outcomes of actions can be probabilistic.

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- § “Stochastic” implies that outcomes of actions can be probabilistic.
- § **Episodic vs. sequential:** In an episodic task environment, the agent’s experience is divided into atomic episodes. The next episode does not depend on the actions taken in previous episodes.
- § For example, an agent spotting defective parts on an assembly line bases each decision on the current part, regardless of previous decisions.
- § In sequential environments, the current decision could affect all future decisions. Chess and taxi driving are sequential in nature.

Properties of Task Environments

- § **Static vs. dynamic:** If the environment can change while an agent is deliberating, then we say the environment is dynamic for that agent; otherwise, it is static.
- § Static environments are easy to deal with because the agent need not keep looking at the world while it is deciding on an action.
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- § Taxi driving is dynamic: the other cars and the taxi itself keep moving while the driving algorithm dithers about what to do next.
- § **Discrete vs. continuous:** This applies to the state of the environment, to the way time is handled, and to the percepts and actions of the agent.
- § Chess has a finite number of distinct states, a discrete set of percepts and actions.
- § the speed and location of the taxi sweep through a range of continuous values and do so smoothly over time. Taxi-driving actions are also continuous (steering angles, *etc.*)

The Structure of Agents

- § Remember that **agent function** maps any percept sequence to an action.
- § Technically, the actual implementation of an agent function running within a physical system is an **agent program**.
- § Another important difference is: agent program takes the current percept as input, and the agent function takes the entire percept history.
- § Four basic kinds of agent programs.
 - ▶ Simple reflex agents.
 - ▶ Model-based reflex agents.
 - ▶ Goal-based agents.
 - ▶ Utility-based agents.

Simple Reflex Agents

- § These agents select actions on the basis of the current percept, ignoring the rest of the percept history.
- § For example, the decision of a vacuum agent is based only on the current location and on whether that location contains dirt.
- § These are the condition–action rule or if-then rule based agents.

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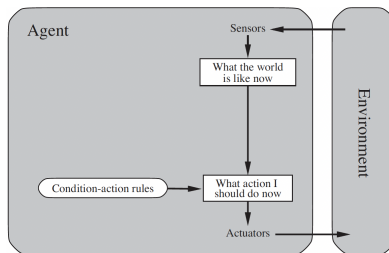


Image Credit: AIMA by Russell and Norvig

- § Simple reflex agents are 'simple', but are of limited intelligence.
- § Works if the correct decision can be made on the basis of only the current percept (fully observable).

Model-based Reflex Agents

- § Partial observability is handled by keeping track of the part of the world it can't see now.
- § The agent should maintain some sort of internal state that depends on the percept history and thereby reflects at least some of the unobserved aspects of the current state.
- § For driving tasks such as changing lanes, the agent needs to keep track of where the other cars are if it can't see them all at once.

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- § For driving tasks such as changing lanes, the agent needs to keep track of where the other cars are if it can't see them all at once.
- § Updating this internal state requires two kinds of knowledge.
 - ▶ Some information about how the world evolves independently of the agent, e.g., an overtaking car generally will be closer behind than it was a moment ago.
 - ▶ Some information about how the agent's own actions affect the world e.g., turning the steering wheel clockwise causes the car turns to the right.
- § This knowledge about “how the world works” is called a model of the world - thus a model-based agent.

Model-based Reflex Agents

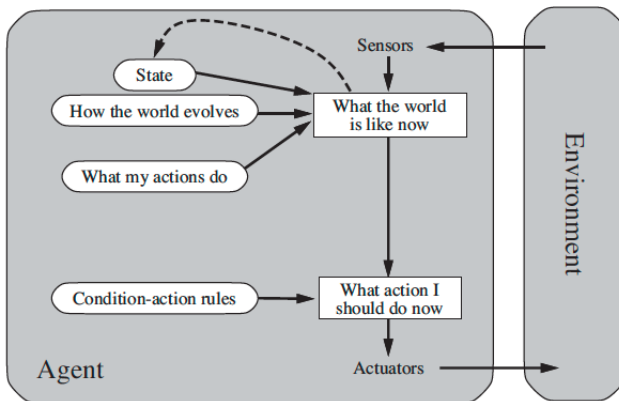


Image Credit: AIMA by Russell and Norvig

Goal-based Agents

- § Knowing something about the current state of the environment is not always enough to decide what to do.
- § For example, the correct decision for a Taxi at a junction depends on where the taxi is trying to get to.
- § The agent needs some sort of goal information that describes situations that are desirable.
- § The agent program can combine this with the model to choose actions that achieve the goal.
- § Search (will be discussed next) and planning are the subfields of AI devoted to finding action sequences that achieve the agent's goals.

Utility-based Agents

- § Goals alone are not enough to generate high-quality behavior in most environments.
- § For example, many action sequences will get the taxi to its goal but some are quicker, safer, more reliable, or cheaper than others.
- § Goals just provide a crude binary distinction between “happy” and “unhappy” states.
- § A more general performance measure should allow a comparison of different world states according to exactly how happy they would make.
- § Because “happy” does not sound very scientific, economists and computer scientists use the term **utility** instead.
- § A utility function is essentially an internalization of the performance measure.

Utility-based Agents

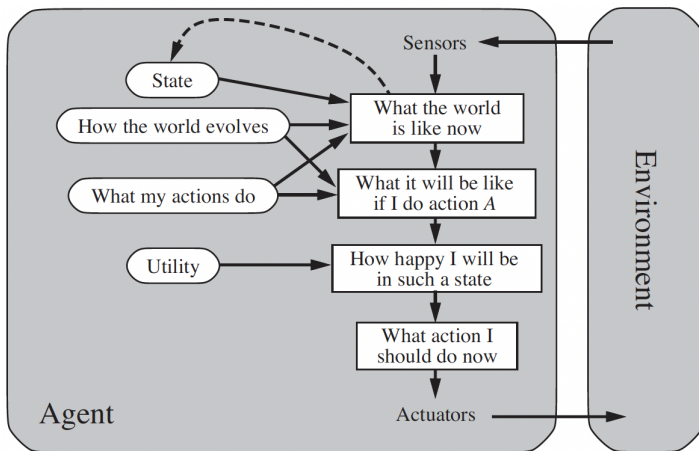


Image Credit: AIMA by Russell and Norvig

How do represent the current state

- § Representations with increasing complexity and expressive power - atomic, factored and structured.
- § In an atomic representation, each state of the world is indivisible. Its a “black box” whose only discernible property is that of being identical to or different from another black box.
- § A factored representation splits up each state into a fixed set of variables or attributes, each of which can have a value.
- § For many purposes, we need to understand the world as having things in it that are related to each other, not just variables with values. This is what constitutes a structured representation.

